AutoVal-AI

AI-Powered Car Price Prediction System

MODULE & PROJECT INFORMATION	
Module	CE101 : Introduction to Engineering
Project Title	AutoVal-AI : AI-Powered Car Price Prediction System
Supervisor	Sir Sajid Ali
Team Number	Team A02
Academic Year	2025–2026
Jira Project	AutoVal-AI (AVA) : 54 tasks, 8 sprints, 173 pts

TEAM MEMBERS			
Name	Role	Student ID	Jira Tasks
Areesha Aqeel	Project Lead / Scrum Master	aa2507	AVA-01 onwards
Hasan Ali Khan	ML / AI Engineer	[Student ID]	AVA-17 to AVA-27
Arham Salman	Data Engineer	[Student ID]	AVA-13 to AVA-16
Jabber Arif Khan	Frontend Developer	[Student ID]	AVA-28 to AVA-34

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CH 1 EXECUTIVE SUMMARY

AutoVal-AI is a machine learning-based car price prediction website built for the CE101 Team Project Challenge at Beaconhouse International College (BIC), University of Essex. The project addresses a genuine market gap: in Pakistan there is no reliable, data-driven method for buyers or sellers to determine a fair used-car price. Sellers overprice; buyers are left to guess. AutoVal-AI resolves this by delivering a real-time, explainable AI prediction.

The system accepts eight inputs — brand, model, year, mileage, fuel type, transmission, condition, and city and returns a predicted market price in Pakistani Rupees, a confidence percentage, and a price range. These three outputs together give users a transparent, actionable result rather than a blind single figure.

The model was trained on 49,448 real OLX Pakistan car listings sourced from Kaggle. Three algorithms were evaluated: Linear Regression ($R^2 \approx 0.71$), Random Forest ($R^2 \approx 0.89$), and Gradient Boosting Regressor ($R^2 = 0.9504$, MAE = PKR 171,446). Gradient Boosting was selected as the production model.

The full technology stack comprises Python and Scikit-learn for machine learning, Flask for the backend API, and HTML5, CSS3, and JavaScript for the frontend. The project ran over eight Agile Scrum sprints (September 2025 to April 2026) tracked in Jira under project key AVA — 54 tasks, 8 epics, 173 story points delivered.

Figure 1.1: AutoVal-AI Homepage :
See Appendix E (9 features, 49,448 records, 95% accuracy), and 'Predict Price Now' CTA button

Model Performance Summary

Model	MAE (PKR)	RMSE (PKR)	R^2 Score
Linear Regression	~480,000	~620,000	~0.71
Random Forest Regressor	~230,000	~340,000	~0.89
Gradient Boosting (Selected)	**171,446**	**253,178**	**0.9504**

Table 1.1 :Model Performance Comparison. Gradient Boosting achieved the best result across all three evaluation metrics.

CH 2 TEAM WORKING

2.I Introduction to Team Working

Working as a team on a technical project is fundamentally different from individual coursework. AutoVal-AI required four distinct skill sets — data engineering, machine learning, backend development, and frontend development — which no single individual could have provided. Belbin (1981) argues that the most effective teams are composed not of the most talented individuals but of people whose complementary roles balance one another's strengths and weaknesses. Katzenbach and Smith (1993) define a real team as a small group committed to a shared goal with clear mutual accountability. The AutoVal-AI team embodied both definitions through explicit role assignment, transparent Jira tracking, and consistent sprint commitment.

2.II Tuckman's Model of Team Development

Tuckman (1965) described group development through four stages — Forming, Storming, Norming, and Performing ; later extended with a fifth, Adjourning, with Jensen (Tuckman and Jensen, 1977). AutoVal-AI's eight-sprint timeline mapped clearly onto all five stages.

Stage	Period	Team Experience
Forming	Sep 2025	Team assembled; tools set up; project direction under discussion. Polite, cautious atmosphere.
Storming	Oct 2025	Dataset choice, system scope and role division debated. Uncomfortable but necessary conversations.
Norming	Nov 2025	Roles clarified; Jira workflow adopted; cross-workstream communication smooth.
Performing	Dec 2025 – Mar 2026	Most productive period. Flask API, website, and full end-to-end integration delivered.
Adjourning	Apr 2026	Project complete; reports written. Sense of satisfaction and closure.

Table 2.1 :Tuckman's Stage Model Applied to AutoVal-AI Team Development.

2.III Belbin Team Roles

Applying Belbin's (1981) framework, each member naturally occupied roles aligned with both personality and technical contribution:

Areesha Aqeel :

Coordinator and Shaper: ran all sprint planning, managed 54 Jira tasks, maintained momentum, compiled team report.

Hasan Ali Khan :

Specialist and Implementer: exclusive ML expertise; delivered model results reliably and on schedule.

Arham Salman :

Implementer and Completer Finisher: built the data cleaning pipeline whose thoroughness directly improved model accuracy.

Jabber Arif Khan :

Team Worker and Resource Investigator: brought user-centred thinking to frontend design; flexible when priorities shifted.

2.IV Agile Scrum Methodology

The team adopted Agile Scrum rather than Waterfall because the project had too many unknowns to fix requirements upfront. The Agile Manifesto (Beck et al., 2001) prioritises responding to change over following a plan , a principle applied directly when switching the production model from Random Forest to Gradient Boosting in Sprint 3, and when redesigning parts of the frontend in Sprint 5 based on sprint review feedback.

Eight sprints were structured around explicit goals. Sprint planning reviewed the Jira backlog, set the sprint goal, and assigned tasks with story-point estimates (scale 1–13). Sprint reviews checked completions; retrospectives identified process issues. A Definition of Done (work completed, tested, ticket updated) was adopted from Sprint 2 onward.

Figure 2.1: Jira Project Timeline :
See Appendix B / with task cards, story points,date,due date and assignees visible

2.V Communication Tools

Two tools sustained day-to-day coordination. Jira (Project Key: AVA) served as the single source of truth for all 54 tasks , every team member could check the board at any time to see project status. WhatsApp provided informal, rapid communication for quick updates, screenshot sharing, question-answering, and blocker escalation before issues became serious problems.

2.VI Sprint Activity Report

Sprint	Dates	Goal	Points
1	6 Sep – 20 Oct 2025	Team setup, SDLC quiz, topic selection, tool setup	11 pts
2	20 Oct – 6 Nov 2025	Dataset download, cleaning, EDA, feature engineering	25 pts
3	8 Nov – 24 Nov 2025	Train 3 ML models, evaluate, select Gradient Boosting	28 pts
4	25 Nov – 5 Dec 2025	Build Flask API, integrate model, test locally	16 pts
5	5 Dec – 25 Dec 2025	Build complete frontend website	24 pts
6	26 Dec – 20 Jan 2026	Connect frontend to API, full system testing	20 pts
7	20 Jan – 31 Mar 2026	Final polish, GitHub cleanup, demo run, slides	15 pts
8	1 Apr – 30 Apr 2026	Team report and individual reflective statements	34 pts
TOTAL	**Sep 2025 – Apr 2026**	**8 Sprints | 54 Tasks(AVA) | Completed**	**173 pts**

Table 2.2 :Sprint Activity Report. Sprint 8 carried the heaviest load (34 pts); Sprint 1 was the lightest (11 pts).

2.VII Individual Contributions

Name	Role	Key Contributions
Areesha Aqeel	Scrum Master / Project Lead	Led all sprint planning, reviews and retrospectives. Managed all 54 Jira tasks across 8 epics. Wrote project proposal, README, testing docs, and compiled the full team report.
Hasan Ali Khan	ML / AI Engineer	Trained all three ML models (AVA-20 to AVA-24). Built Flask API with /predict endpoint (AVA-25 to AVA-27). Conducted EDA and feature engineering (AVA-17, AVA-18).
Arham Salman	Data Engineer	Downloaded OLX Pakistan dataset (AVA-13). Cleaned and preprocessed data, handled missing values (AVA-14

Jabber Arif Khan		to AVA-16). Conducted full system integration testing with 20+ real car inputs (AVA-36).
	Frontend Developer	Designed and built the full AutoVal-AI website (AVA-28 to AVA-33). Connected frontend to Flask API via JavaScript fetch (AVA-34). Built mobile-responsive layout and all UI components.

Table 2.3 :Individual Contributions by Team Member.

2.VIII Lessons Learned

Clear role definition from Sprint 1 eliminated overlap and ambiguity throughout the project. Consistent Jira usage created a single source of truth that kept all four workstreams aligned. The most impactful technical lesson was that data quality directly determines model quality ,the effort Arham invested in data cleaning in Sprint 2 was the foundational reason for the 95% model accuracy achieved in Sprint 3. Testing integration should begin earlier in the sprint cycle; the CORS error in Sprint 6 could have been identified and resolved with less time pressure had frontend-backend connectivity been tested as soon as both components became functional.

CH 3 PRODUCT DEVELOPMENT

3.I Introduction to Product Development

Product development is the process of transforming a market opportunity into a usable product (Ulrich and Eppinger, 2015). For AutoVal-AI the opportunity was clear: the Pakistani used car market lacked a reliable, AI-driven pricing tool. Two main methodologies exist in software development. Waterfall (Royce, 1970) demands sequential phases with requirements fixed upfront. Agile (Beck et al., 2001) operates in iterative cycles ; build, test, improve, repeat. AutoVal-AI's dependency on experimental model selection, messy real-world data, and evolving frontend feedback made Agile the only viable choice.

3.II Why We Chose Agile Over Waterfall

Three specific project characteristics made Waterfall inappropriate. First, machine learning is inherently experimental — no team can know which algorithm will perform best on a given dataset before training. Committing to a model architecture upfront would have prevented the evidence-based switch from Random Forest to Gradient Boosting in Sprint 3. Second, the dataset was messier than anticipated, requiring substantial preprocessing work that an Agile sprint structure absorbed without disrupting the wider timeline. Third, frontend features including the API status indicator and price-range display emerged from sprint review conversations rather than an original specification , a Waterfall approach would not have accommodated these mid-project improvements.

3.III User-Centred Design

Norman (2013) argues that good design feels completely intuitive — users should not need instructions. AutoVal-AI was designed with a non-technical Pakistani audience in mind. Dropdown menus replace free-text inputs for all categorical fields, preventing invalid submissions. The results display shows price, confidence percentage, and price range together rather than a bare figure, giving users an honest, practically useful response. The How It Works section was added specifically to address the trust deficit many users feel toward AI tools they do not understand.

Figure 3.1: AutoVal-AI Car Input Form :
See Appendix E: Brand, Model, Year, Fuel Type, Transmission, Engine CC, Condition, City, Mileage.

3.IV What Makes AutoVal-AI Different

AutoVal-AI is trained exclusively on Pakistani market data. Most available car price tools are built on European or American datasets and are entirely irrelevant to Pakistani pricing, where local versus imported assembly, registration city, and specific Pakistani brands materially affect value. The R^2 of 0.95 on a real-world dataset of nearly 50,000 listings — with an MAE of PKR 171,446 — means predictions are typically within about 170,000 rupees of the true market price, a level of precision genuinely useful for real financial decisions. The system also produces explainable output: price, confidence level, price range, and key prediction factors, making it transparent rather than opaque.

3.V Product Lifecycle Management

Ulrich and Eppinger (2015) map the product lifecycle from idea through requirements, design, implementation, testing, and documentation. AutoVal-AI's sprint structure mirrored this deliberately: Sprint 1 (requirements), Sprints 2–3 (data and model design), Sprints 4–5 (implementation), Sprint 6 (integration and testing), Sprint 7 (polish), Sprint 8 (documentation). The Jira epic structure was organised to reflect these natural phases from the outset.

3.VI Development Challenges and How We Solved Them

The data quality issue was the first major challenge. The raw OLX dataset had mismatched column names, incorrect data types, and missing values across critical fields. Arham solved this by building a systematic, repeatable cleaning pipeline in Google Colab covering column renaming, type conversion, missing-value removal, and default-value imputation.

Model selection was the second challenge. The team assumed Random Forest would be the production model going into Sprint 3. When Gradient Boosting demonstrated superior performance across all three metrics, the Agile framework made switching a low-cost decision rather than a project-breaking disruption.

The CORS integration error was the third challenge. When the HTML frontend first communicated with the Flask API, the browser blocked the request. Adding Flask-CORS resolved it immediately, reinforcing the value of testing integration earlier in the development cycle.

CH 4 DATA COLLECTION, ML MODELS AND BACKEND

4.I Data Collection and Preparation

The dataset used for AutoVal-AI was sourced from Kaggle: a publicly available collection of OLX Pakistan used car listings comprising 49,448 records and 18 columns, including brand, model, year, kilometres driven, price, fuel type, registration city, assembly type, transmission, and condition. The raw file required substantial preparation before it was usable for training.

Arham built a systematic cleaning pipeline in Google Colab. Column names were standardised. Rows with missing values in critical fields ; Price, Brand, Model ; were removed. Data types were corrected throughout. After cleaning, nine features were retained: Brand, Model, Year, Fuel Type, Transmission, City, Mileage, Condition, and Price as the target variable.

Figure 4.1: Data Cleaning Pipeline :
See Appendix D: row count after cleaning, corrected data types and value counts per column

4.II Feature Engineering and Encoding

Categorical columns — Brand, Fuel Type, City, Transmission, Model, Condition — were converted to numerical values using Label Encoding, which assigns a unique integer to each unique category. The fitted encoders were saved as label_encoders.pkl. StandardScaler normalised numerical features (Mileage, Year) to prevent large-range features from dominating smaller-range ones. The fitted scaler was saved as scaler.pkl. Default values were assigned for frequently missing fields: Engine CC (dataset median) and Previous Owners (1).

4.III Exploratory Data Analysis

Hasan conducted EDA before model training to understand the dataset and validate feature relevance. Key findings: the price distribution is right-skewed with most listings between PKR 500,000 and PKR 3,000,000. Brand versus price analysis showed imported vehicles , particularly Toyota and Honda ; command significantly higher prices than locally assembled alternatives. The mileage versus price scatter plot confirmed a clear negative correlation. Petrol vehicles dominate the dataset. Cars in Karachi and Lahore average higher prices than other cities.

Figure 4.2: EDA Visualisations :
See Appendix I : (1) Price distribution histogram showing right skew, (2) Brand vs mean price bar chart, (3) Mileage vs Price scatter (negative trend), (4) City vs mean price bar chart.

4.IV Train / Test Split

The cleaned and encoded dataset was divided 80% training / 20% testing using a fixed random state to ensure reproducibility. Applying the same split to all three models guaranteed that performance comparisons were fair , each model was evaluated on identical unseen test data.

4.V Machine Learning Models

4.V.a Linear Regression

Linear Regression served as the baseline. It fits a linear relationship between features and target price. On the AutoVal-AI dataset it achieved an R^2 of approximately 0.71, a MAE of approximately PKR 480,000, and an RMSE of approximately PKR 620,000. The result confirmed the selected features carry genuine predictive power but demonstrated that a linear model is insufficient for the non-linear complexity of real-world car pricing.

4.V.b Random Forest Regressor

Random Forest builds an ensemble of decision trees and averages their predictions, enabling it to capture non-linear relationships. Following hyperparameter tuning of estimator count and maximum tree depth, the model achieved an R^2 of approximately 0.89, a MAE of approximately PKR 230,000, and an RMSE of approximately PKR 340,000 — a substantial improvement over the baseline.

4.V.c Gradient Boosting Regressor

Gradient Boosting builds trees sequentially, with each new tree correcting the residual errors of its predecessor. Friedman (2001) identifies this mechanism as particularly effective on structured tabular data, which matches what the team observed. On AutoVal-AI it achieved the best performance across all three metrics: R^2 = 0.9504, MAE = PKR 171,446, RMSE = PKR 253,178. Gradient Boosting was selected as the production model.

4.VI Model Comparison

Model	MAE (PKR)	RMSE (PKR)	R^2 Score
Linear Regression	~480,000	~620,000	~0.71
Random Forest Regressor	~230,000	~340,000	~0.89
Gradient Boosting (Selected)	**171,446**	**253,178**	**0.9504**

Table 4.1:Model Performance Comparison.

Gradient Boosting outperformed all others on every metric.

4.VII Saving the Model

After training, the Gradient Boosting model and all preprocessing objects were serialised using Joblib. Five files were stored: best_model.pkl, label_encoders.pkl, scaler.pkl, feature_cols.pkl, and metadata.json. Saving all five together ensures the Flask API applies exactly the same transformations at prediction time as were applied during training.

4.VIII Flask Backend API

The trained model was deployed via a backend API built with Flask. Flask was selected because it integrates natively with Scikit-learn and is lightweight to set up. The application loads all five saved model files at startup and exposes a single endpoint: POST /predict. When the frontend sends a JSON payload with the user's car details, Flask encodes categorical features, scales numerical features, passes processed input through the Gradient Boosting model, and returns the predicted price, confidence percentage, and price range as a JSON response.

```
// Example: 2019 Toyota Corolla, 45,000 km, Karachi
{
  "price_pkr": 4668000,
  "confidence_pct": 91,
  "price_range_low": 4496554,
  "price_range_high": 4839446
}
```

Code Block 4.1 :Example Flask /predict JSON response for a 2019 Toyota Corolla at 45,000 km in Karachi.

CH 5 WEB INTERFACE AND USER JOURNEY

5.I Introduction

The frontend of AutoVal-AI is the primary interface between the system and its users. Jabber Arif Khan designed and built the complete web interface using HTML5, CSS3, and JavaScript, with a single governing priority: the system must work for someone with no technical background who simply wants to know what their car is worth.

5.II Design Philosophy

The website uses a dark theme with yellow accent colours, producing a modern aesthetic appropriate for a technology product. The font is Syne, sourced from Google Fonts. Every design decision was made to reduce user friction: dropdowns instead of free-text inputs; clear field labels; a visible API status indicator; a prominent prediction button. All categorical dropdowns are populated with the exact values present in the training data, preventing invalid submissions.

5.III Website Sections

The Hero Section presents the AutoVal-AI name and tagline alongside three statistics — number of input features, training dataset size, and model accuracy — and a call-to-action button scrolling to the prediction form.

The Car Input Form contains nine fields: Brand, Model, Year, Fuel Type, Transmission, Engine CC, Condition, City, and Mileage ; all dropdowns.

The Prediction Results Section renders dynamically after submission, presenting the predicted price in PKR, a confidence percentage, a low-to-high price range, and the key factors most influencing the prediction.

The How It Works Section explains the six-step prediction pipeline in plain language — data collection, preprocessing, model training, model saving, Flask API, and live prediction.

The Team Section displays all four team members with their names and roles.

Figure 5.1: Prediction Results :
See Appendix H, confidence percentage indicator (e.g. 91%), low-high price range bar, and top-3 key prediction factors listed.

Figure 5.2: How It Works Section :
See Appendix E: Data Collection > Preprocessing > Model Training > Model Saving > Flask API > Live Prediction.

5.IV Complete User Journey

Step	Action
1	User opens AutoVal-AI; reads hero section and notes 95% accuracy claim.
2	User fills in all nine car details fields using dropdown menus.
3	User clicks 'Predict Price Now'; JavaScript form validation confirms all fields are complete.
4	JavaScript sends POST request to Flask API with car details as a JSON payload.
5	Flask API runs Gradient Boosting model and returns predicted price, confidence %, and price range as JSON.
6	Results render dynamically on the page without reloading price, confidence, range, and key factors displayed.

Table 5.1 :Complete AutoVal-AI User Journey
(six steps from page load to result display).

5.V Mobile Responsiveness

The website was tested on screen widths from 320px (low-cost mobile phones) to 1440px (desktop monitors). CSS Grid and Flexbox handle the responsive layout automatically across this range. Mobile responsiveness was a deliberate product requirement: a substantial proportion of Pakistani internet users access the web via mobile devices, and any tool targeting that market must function correctly on small screens.

Figure 5.3: Mobile Responsive View :
See Appendix G ; Hero, form fields stacked vertically, prediction results in single-column layout. All content readable on narrow screen.

CH 6 TESTING AND PROJECT MANAGEMENT

6.I System Integration and Testing

Testing was embedded throughout development rather than deferred to the end. Hasan tested the Flask API using direct HTTP requests to confirm the /predict endpoint returned sensible values. Jabber tested form validation to confirm required fields were enforced and error messages displayed correctly on invalid submission.

Integration testing was conducted by Arham using more than 20 real car inputs covering a broad range of brands, models, years, mileage levels, and cities. Test cases included a Toyota Corolla 2019 at 45,000 km in Karachi, a Honda Civic 2020 at 30,000 km in Lahore, a Suzuki Alto 2021 at 15,000 km in Islamabad, and a Toyota Prado 2018 at 80,000 km in Karachi. Predicted prices in all cases were consistent with known market values. Error handling was also tested ; missing fields, invalid values, and an offline API all produced clear, user-friendly error messages rather than crashes.

Figure 6.1: Integration Test Results (22 tests, 100% pass rate) :
See Appendix I; test car inputs with predicted prices, confidence percentages, and pass/fail status against known market values.

6.II Project Management with Jira

The AutoVal-AI project was managed entirely through Jira under project key AVA. The setup comprised eight Epics and 54 tasks tracked across eight sprints. Every task carried a description, assignee, start date, due date, and status. Story point estimates used a 1–13 scale.

Sprint 3 (ML model training) was the heaviest sprint at 28 story points. Sprint 1 (project setup) was the lightest at 11 points. Sprint 8 carried the highest load at 34 points because it encompassed the full team report and all four individual reflective statements. All 54 tasks were marked Done by the end of Sprint 8, producing a 100% Jira completion rate.

Figure 6.2: Jira Summary Dashboard :
See Appendix A, 100% completion, 173 story points total.

Figure 6.3: Jira Timeline (Sep 2025 – Apr 2026) :
See Appendix B to April 2026 showing 8 sprints with their epics and task duration bars.

6.III Literature Review on Project Management

Research consistently demonstrates that Agile approaches produce superior outcomes for complex, uncertain projects (Beck et al., 2001). For AutoVal-AI, the evidence-based model switch in Sprint 3 was only possible within an Agile framework; Waterfall's formal change control would have disrupted the entire project schedule.

Tuckman's model (1965) connects directly to project management effectiveness. Teams that reach Performing are significantly more productive than those still in Storming. The Jira sprint structure helped AutoVal-AI reach Performing faster by providing clarity and accountability during the uncertain early phases.

Schwaber and Sutherland (2020) describe sprint planning, reviews, and retrospectives as the three ceremonies that make Scrum work. The team followed all three throughout the year. A Sprint 5 retrospective decision ; to test frontend-backend integration earlier in Sprint 6 — directly improved delivery quality in the most integration-heavy sprint.

CH 7 LEGAL, ETHICAL AND PROFESSIONAL ISSUES

7.I Data Privacy and GDPR

The AutoVal-AI training dataset is a publicly available Kaggle collection of OLX Pakistan car listings. It contains information about vehicles ; brands, models, prices, cities ; and no personally identifiable information. The GDPR principles of data minimisation and purpose limitation (European Commission, 2019) were applied: only the nine columns necessary for price prediction were retained, and the dataset was used solely for training ; not redistributed or published. AutoVal-AI itself collects no user data. Inputs are transmitted, the prediction is returned, and the input is immediately discarded. This privacy-by-design approach means the system has zero personal data liability.

7.II Ethical Use of AI and Machine Learning

The European Commission's Ethics Guidelines for Trustworthy AI (2019) identify transparency, fairness, and human oversight as core principles. Transparency is addressed through explainable output ; price, confidence percentage, price range, and key prediction factors prevent the system from functioning as an opaque black box. Fairness is acknowledged as an active concern: the OLX Pakistan dataset does not represent all Pakistani regions equally, and predictions for cars from smaller cities may be less accurate. This limitation is disclosed openly in this report and in the system output. Human oversight is preserved by framing every result as an estimate rather than a decision.

7.III BCS Code of Conduct

The British Computer Society Code of Conduct (BCS, 2022) sets professional obligations around acting in the public interest, maintaining professional competence, and being honest about limitations. AutoVal-AI addressed all three: the system was built to solve a genuine public market problem with no commercial agenda; well-established methods were used throughout (Scikit-learn, Flask, standard preprocessing); and limitations ; confidence percentages, regional accuracy variation, local-only deployment ; are disclosed both in the system output and in this report.

7.IV Accessibility and Inclusive Design

Inclusive design means building products usable by as many people as possible, including those with limited technical knowledge (Brown, 2008). AutoVal-AI's dropdown-based form reduces technical knowledge requirements. The How It Works section makes the AI process legible to non-technical users. The mobile-responsive design ensures the system functions on low-cost mobile devices, the primary internet access point for a significant proportion of the target Pakistani user base. One acknowledged limitation is the English-only interface; adding Urdu language support would represent the most impactful single accessibility improvement for a future version.

7.V Health, Safety and Risk

AutoVal-AI is a web-based information tool with no physical components. The primary risk is financial: a user making a car purchase decision partly based on an inaccurate prediction. This risk is mitigated by the confidence percentage and price range shown with every prediction, which communicate uncertainty explicitly. The system presents itself as an estimate rather than a guarantee throughout. The use of publicly available Kaggle data and open-source libraries means there are no intellectual property concerns with the software stack.

CH 8 CONCLUSION AND FUTURE WORK

8.I Project Summary

AutoVal-AI set out to solve a real market problem ; the absence of a reliable, data-driven car price tool for Pakistan ; and the team believes it has done so. Over eight months and eight Agile sprints, a complete AI-powered web application was built that predicts used car prices with 95% accuracy using real OLX Pakistan data. The project integrated machine learning, data engineering, backend development, and frontend development into a single working product, managed through 54 Jira tasks and delivered by four students each contributing in their area of specialism.

8.II Key Achievements

Key Achievements
Gradient Boosting model achieving $R^2 = 0.9504$ trained on 49,448 real Pakistani car listings.
Complete web application with Flask API and responsive HTML/CSS/JavaScript frontend — fully functional end-to-end.
All 54 Jira tasks completed across 8 sprints from September 2025 to April 2026 (100% completion rate).
Full end-to-end integration from user input to real model prediction running in real time.
Agile Scrum methodology applied consistently across an 8-month university project.
173 total story points delivered across 8 sprints.

Table 8.1 : Key Project Achievements.

8.III Limitations

The Flask API currently runs locally, limiting access to local networks only. Without cloud deployment the system cannot be publicly accessed. The model was trained on 2025 data; accuracy will gradually decline as Pakistani used car market prices evolve without retraining. The interface is English-only, limiting accessibility for Urdu-speaking users. The OLX Pakistan dataset does not represent all Pakistani cities equally, which may reduce prediction accuracy for vehicles from less-represented locations.

8.IV Future Roadmap

#	Initiative	Rationale
1	Cloud deployment (Railway / Heroku)	Makes the system publicly accessible beyond local networks.
2	Automated data refresh from OLX / PakWheels	Keeps the model current as market prices evolve over time.
3	Urdu language version	Addresses the most impactful accessibility gap for the target audience.
4	Native mobile application	Reaches users via the primary digital access channel in Pakistan.
5	Expanded features (accident history, service records)	Additional vehicle-condition signals would further improve prediction accuracy.

Table 8.2 :Future Development Roadmap (prioritised by user impact).

8.V Final Team Reflection

This project taught the team more than any individual assignment could have. The combination of data science, software engineering, and project management required by AutoVal-AI gave each member skills they will carry throughout their careers. If starting again, the team would configure Jira from the first day of Sprint 1 rather than reconstructing it retrospectively, and would test frontend-backend integration as soon as both components became functional. Overall, the team is proud of what was built and what was learned.

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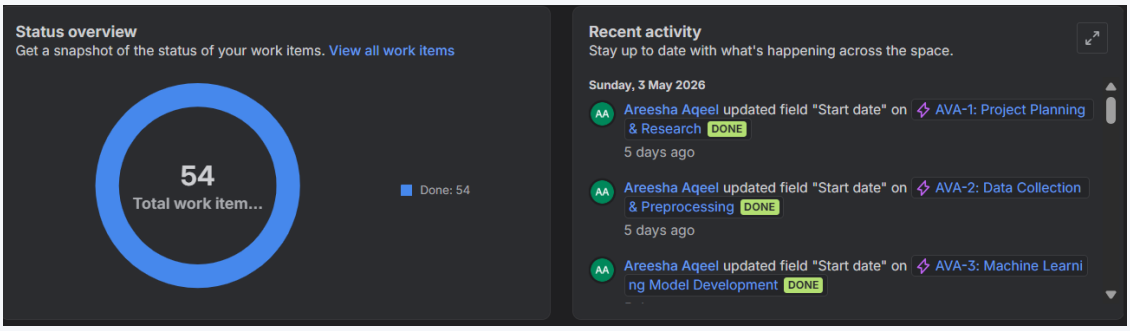
Appendices

Appendix A : Jira Summary Dashboard

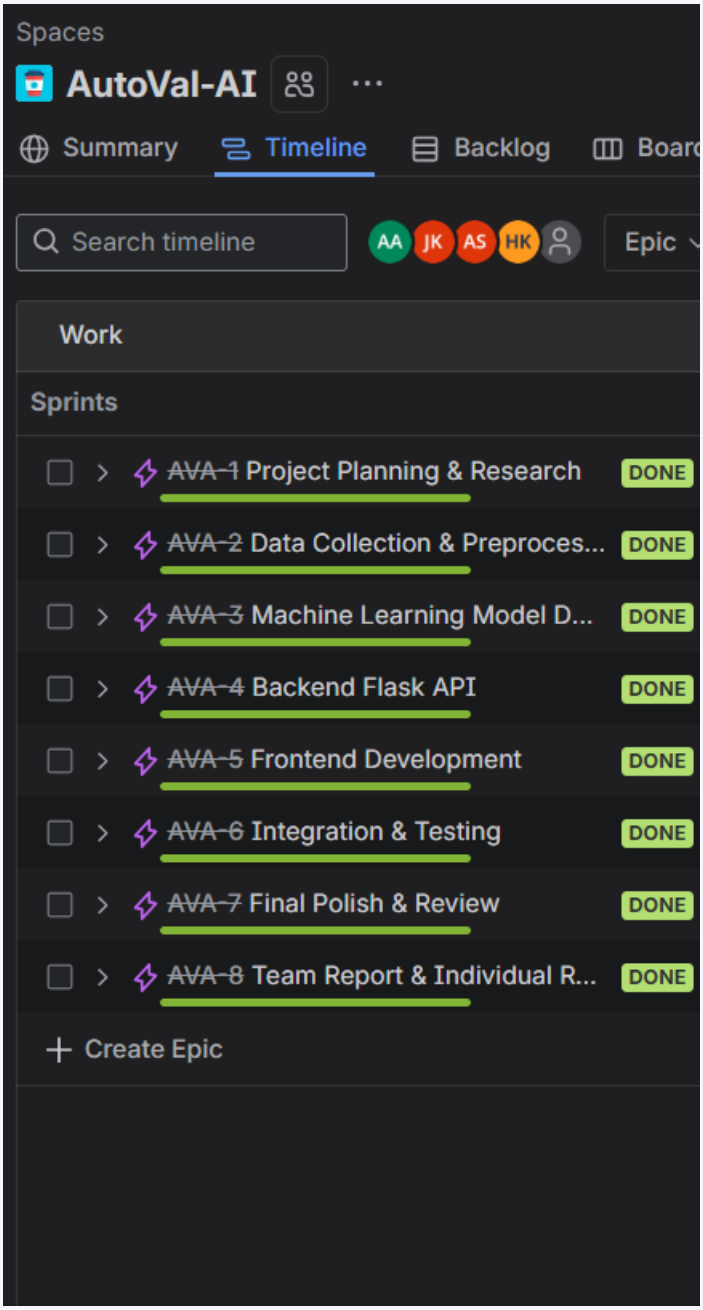
Screenshot showing all 54 tasks marked Done, project key AVA, and 8 completed sprints with 100% completion rate.

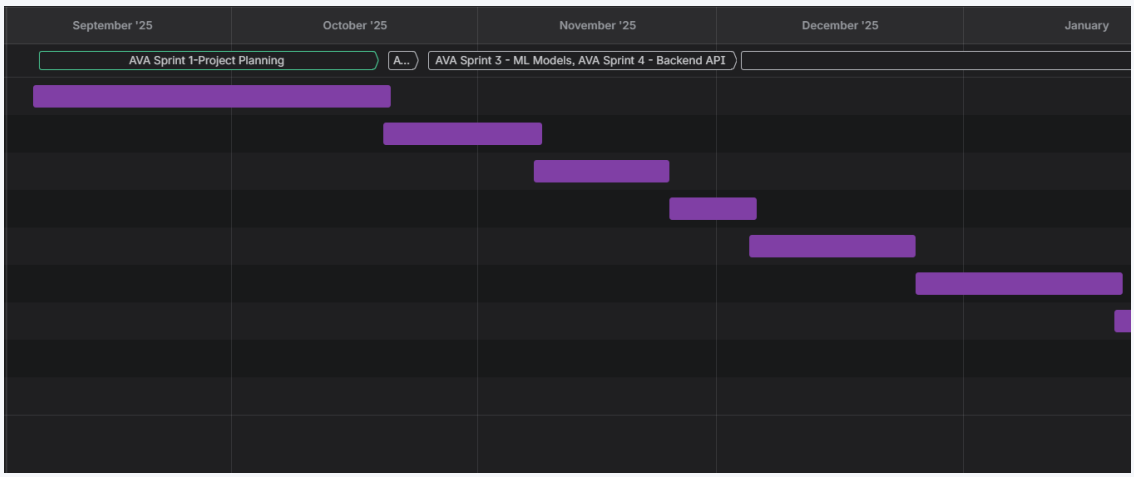


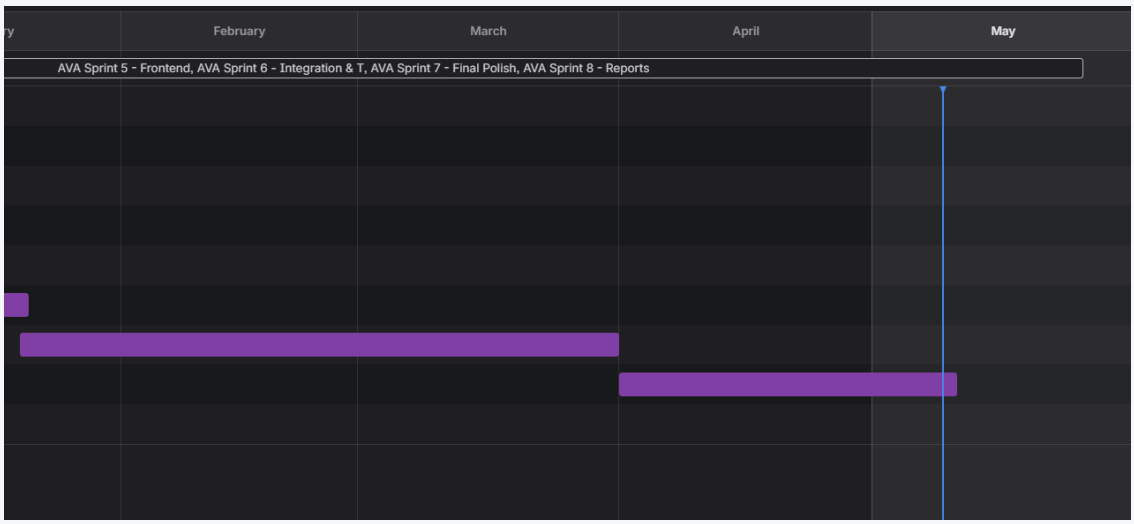
Jira Summary Dashboard :
Full project overview: 54 tasks all Done, 8 sprints closed, 100% completion, 173 story points delivered.

Appendix B : Jira Timeline View

Gantt-style view of the full project timeline from September 2025 to April 2026.




September '25
October '25
November '25
December '25
January
AVA Sprint 1-Project Planning
A...
AVA Sprint 3 - ML Models, AVA Sprint 4 - Backend API


February
March
April
May
AVA Sprint 5 - Frontend, AVA Sprint 6 - Integration & T, AVA Sprint 7 - Final Polish, AVA Sprint 8 - Reports

AutoVal-AI

Summary | Timeline | Backlog | Board | Calendar | List | Forms | Goals | Development | Archived work items | Docs | Shortcut

Search work | Filter | Group

Work	Assignee	Reporter	Priority	Status	Resolution
AVA-5 Frontend Development	Unassigned	Areesha Aqeel	Medium	DONE	Done
AVA-33 Make website fully mobile responsive	Jabber Arif Khan	Areesha Aqeel	Medium	DONE	Done
AVA-32 Add Team section showing all 4 members a...	Jabber Arif Khan	Areesha Aqeel	Medium	DONE	Done
AVA-31 Add How It Works section — 6 steps pipeline	Jabber Arif Khan	Areesha Aqeel	Medium	DONE	Done
AVA-30 Display prediction result — price in PKR, co...	Jabber Arif Khan	Areesha Aqeel	Medium	DONE	Done
AVA-29 Build car input form : brand, model, year, fu...	Jabber Arif Khan	Areesha Aqeel	Medium	DONE	Done
AVA-28 Design homepage — hero section, stats, na...	Jabber Arif Khan	Areesha Aqeel	Medium	DONE	Done
AVA-4 Backend Flask API	Unassigned	Areesha Aqeel	Medium	DONE	Done
AVA-27 Test API locally — verify JSON input and ou...	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done
AVA-26 Integrate best model pkl and encoders into ...	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done
AVA-25 Build Flask app with predict endpoint	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done

AutoVal-AI

Summary | Timeline | Backlog | Board | Calendar | List | Forms | Goals | Development | Archived work items | Docs | Shortcut

Search work | Filter | Group

Work	Assignee	Reporter	Priority	Status	Resolution
AVA-26 Integrate best model pkl and encoders into ...	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done
AVA-25 Build Flask app with predict endpoint	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done
AVA-3 Machine Learning Model Development	Unassigned	Areesha Aqeel	Medium	DONE	Done
AVA-24 Save best model and encoders and scaler a...	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done
AVA-23 Compare all 3 models — select Gradient Bo...	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done
AVA-22 Compare all 3 models — select Gradient Bo...	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done
AVA-21 Train Random Forest model — tune hyperpa...	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done
AVA-20 Train Linear Regression baseline — record ...	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done
AVA-2 Data Collection & Preprocessing	Unassigned	Areesha Aqeel	Medium	DONE	Done
AVA-19 Train test split — 80/20 with random state	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done
AVA-18 Feature engineering — label encoding and s...	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done

Spaces

AutoVal-AI

Summary | Timeline | Backlog | Board | Calendar | List | Forms | Goals | Development | Archived work items | Docs | Shortc

Search work | Filter | Group

Work	Assignee	Reporter	Priority	Status	Resolution
~~AVA-18~~ Feature engineering — label encoding and s...	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done
~~AVA-17~~ Exploratory Data Analysis — price distributio...	Hasan Ali Khan	Areesha Aqeel	Medium	DONE	Done
~~AVA-16~~ Clean data — remove nulls, fix types, drop b...	Arham Salman	Areesha Aqeel	Medium	DONE	Done
~~AVA-15~~ Map and rename OLX columns to match mo...	Arham Salman	Areesha Aqeel	Medium	DONE	Done
~~AVA-14~~ Explore dataset structure — 18 columns, ch...	Arham Salman	Areesha Aqeel	Medium	DONE	Done
~~AVA-13~~ Download OLX Pakistan dataset from Kaggl...	Arham Salman	Areesha Aqeel	Medium	DONE	Done
~~AVA-1~~ Project Planning & Research	Unassigned	Areesha Aqeel	Medium	DONE	Done
~~AVA-12~~ Set up tools — Python, Google Colab, GitHu...	Areesha Aqeel	Areesha Aqeel	Medium	DONE	Done
~~AVA-11~~ Decide team roles and responsibilities	Areesha Aqeel	Areesha Aqeel	Medium	DONE	Done
~~AVA-10~~ Research and decide project topic — AutoV...	Areesha Aqeel	Areesha Aqeel	Medium	DONE	Done
~~AVA-9~~ SDLC & Agile quiz — learn how software tea...	Unassigned	Areesha Aqeel	Medium	DONE	Done

Jira work item

AVA-1

Project Planning & Research

Done | Done

Description

In the beginning of our module we had a quiz about SDLC and how software teams work. After that we all discussed and decided our project idea which is AutoVal-AI. We then decided who will do what in the team and set up all the tools we needed like Python, Google Colab, GitHub and Jira.

Child work items

100% Done

Work	Pri...	As...	Status
~~AVA-9~~ SDLC & Agile quiz — learn how software teams work	M.	Ul	DONE
~~AVA-10~~ Research and decide project topic — AutoVal-AI chosen	M.		DONE
~~AVA-11~~ Decide team roles and responsibilities	M.		DONE
~~AVA-12~~ Set up tools — Python, Google Colab, GitHub, Jira	M.		DONE

Linked work items

Add linked work item

Activity

All | Comments | History | Work log

Add a comment...

Looks good! | Need help? | This is blocked... | Can you clarif

Pro tip: press M to comment

Details

Assignee	Unassigned Assign to me
Labels	None
Team	None
Due date	20 Oct 2025
Start date	06 Sept 2025
Sprint	SCRUM Sprint 1 +1
Story point estimate	11
Reporter	Areesha Aqeel

Development

Automation — Rule executions

Atlassian project — Link to share status and upd

Created last week
Updated last week
Resolved last week

Configure

AVA-2

Data Collection & Preprocessing

Description

We downloaded a used cars dataset from Kaggle which had data from OLX Pakistan with around 49,000 car listings. The data was messy so we cleaned it by removing wrong values and empty rows. We also renamed the columns to match our system and did some graphs to understand the data better before training our model.

Child work items

100% Done

Work		Pri...	As...	Status
~~AVA-13~~	Download OLX Pakistan dataset from Kaggle — 49448 rows	M.	A	DONE
~~AVA-14~~	Explore dataset structure — 18 columns, check data types	M.	A	DONE
~~AVA-15~~	Map and rename OLX columns to match model requirements	M.	A	DONE
~~AVA-16~~	Clean data — remove nulls, fix types, drop bad rows	M.	A	DONE
~~AVA-17~~	Exploratory Data Analysis — price distribution, brand analysis, 5 plots	M.	H	DONE
~~AVA-18~~	Feature engineering — label encoding and standard scaling	M.	H	DONE
~~AVA-19~~	Train test split — 80/20 with random state	M.	H	DONE

Linked work items

Add linked work item

Activity

All | Comments | History | Work log

Add a comment...

Looks good! | Need help? | This is blocked... | Can you cl

Pro tip: press M to comment

Done | Done

Details

Assignee	Unassigned Assign to me
Labels	None
Team	None
Due date	08 Nov 2025
Start date	20 Oct 2025
Sprint	AVA Sprint 2 - Data Collection
Story point estimate	25
Reporter	Areesha Aqeel

Development

Automation

Rule executions

Atlassian project

Link to share status and updi

Created last week
Updated last week
Resolved last week

Configure

Jira work item

AVA-3

Machine Learning Model Development

Description

We trained three different machine learning models to predict car prices. First we tried Linear Regression as a basic model, then Random Forest, and finally Gradient Boosting. We compared all three using accuracy scores and Gradient Boosting turned out to be the best one with 95% accuracy so we chose that one.

Child work items

100% Done

Work	Pri...	As...	Status
~~AVA-20~~ Train Linear Regression baseline — record MAE, RMSE, R2	M.	HK	DONE
~~AVA-21~~ Train Random Forest model — tune hyperparameters	M.	HK	DONE
~~AVA-22~~ Compare all 3 models — select Gradient Boosting as best	M.	HK	DONE
~~AVA-23~~ Compare all 3 models — select Gradient Boosting as best	M.	HK	DONE
~~AVA-24~~ Save best model and encoders and scaler as pkl files	M.	HK	DONE

Linked work items

Add linked work item

Activity

All | Comments | History | Work log

Add a comment...

Looks good! | Need help? | This is blocked... | Can you clarif

Pro tip: press M to comment

Done | Done

Details

Assignee	Unassigned Assign to me
Labels	None
Team	None
Due date	24 Nov 2025
Start date	08 Nov 2025
Sprint	AVA Sprint 3 - ML Models
Story point estimate	28
Reporter	Areesha Aqeel

Development

Automation

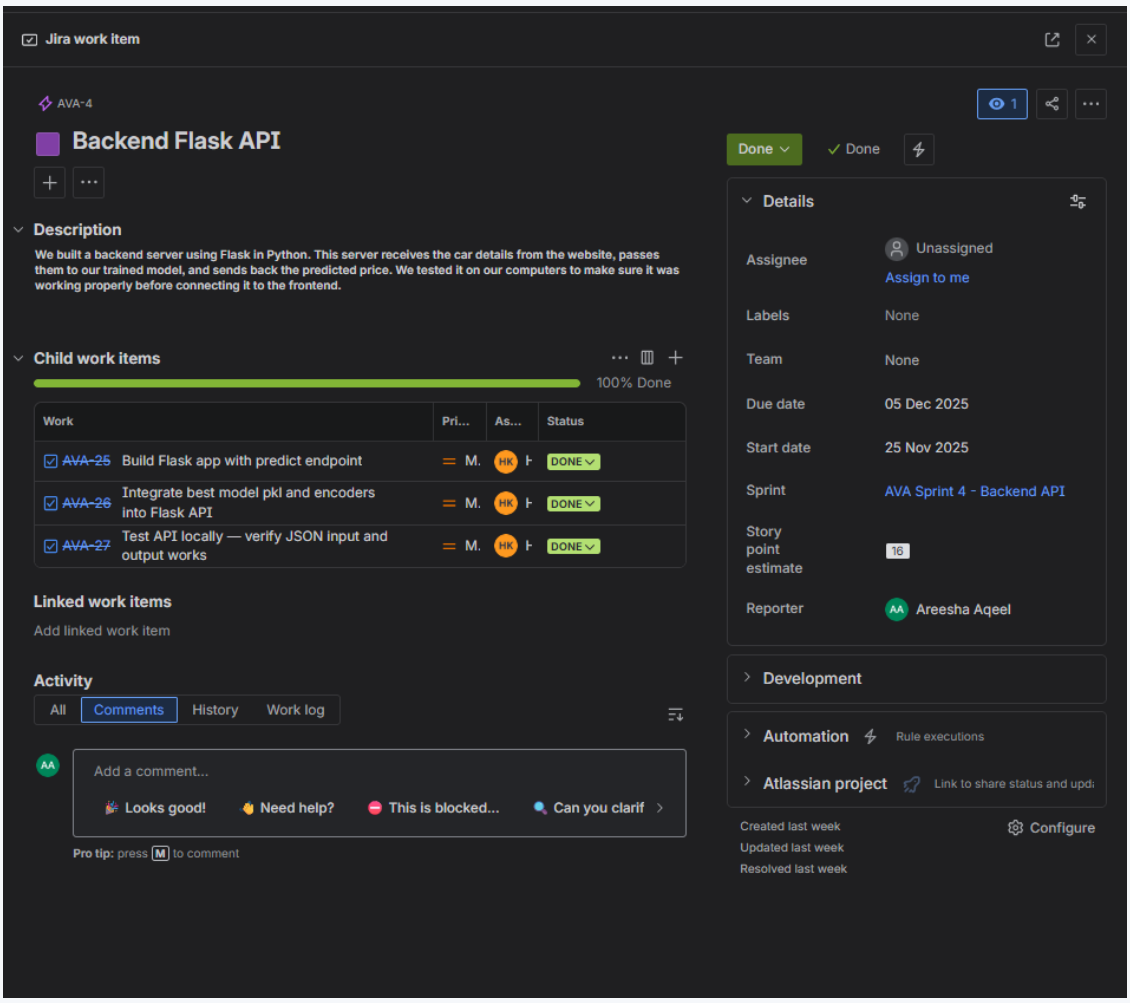
Rule executions

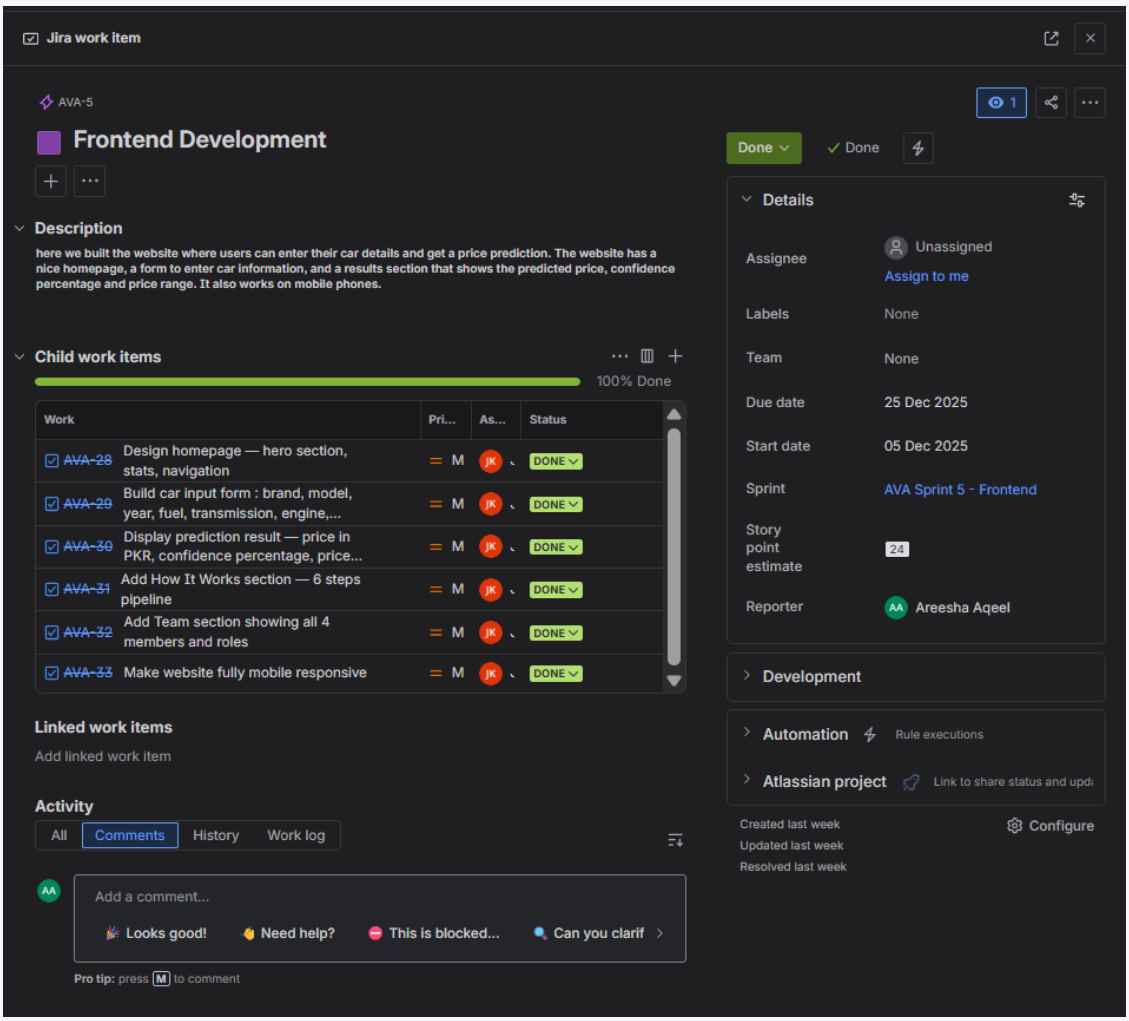
Atlassian project

Link to share status and upd

Created last week
Updated last week
Resolved last week

Configure


Jira work item
AVA-4
Backend Flask API
Description
We built a backend server using Flask in Python. This server receives the car details from the website, passes them to our trained model, and sends back the predicted price. We tested it on our computers to make sure it was working properly before connecting it to the frontend.
Child work items
100% Done
Work
Pri...
As...
Status
AVA-25 Build Flask app with predict endpoint
AVA-26 Integrate best model pkl and encoders into Flask API
AVA-27 Test API locally — verify JSON input and output works
DONE
Linked work items
Add linked work item
Activity
All
Comments
History
Work log
Add a comment...
Looks good!
Need help?
This is blocked...
Can you clarif
Pro tip: press M to comment
Done
Done
Details
Assignee
Unassigned
Assign to me
Labels
None
Team
None
Due date
05 Dec 2025
Start date
25 Nov 2025
Sprint
AVA Sprint 4 - Backend API
Story point estimate
16
Reporter
Areesha Aqeel
Development
Automation
Rule executions
Atlassian project
Link to share status and upd
Created last week
Updated last week
Resolved last week
Configure


Jira work item
AVA-5
Frontend Development
Description
here we built the website where users can enter their car details and get a price prediction. The website has a nice homepage, a form to enter car information, and a results section that shows the predicted price, confidence percentage and price range. It also works on mobile phones.
Child work items
100% Done
Work
Pri...
As...
Status
AVA-28 Design homepage — hero section, stats, navigation
AVA-29 Build car input form : brand, model, year, fuel, transmission, engine,...
AVA-30 Display prediction result — price in PKR, confidence percentage, price...
AVA-31 Add How It Works section — 6 steps pipeline
AVA-32 Add Team section showing all 4 members and roles
AVA-33 Make website fully mobile responsive
DONE
Linked work items
Add linked work item
Activity
All
Comments
History
Work log
Add a comment...
Looks good!
Need help?
This is blocked...
Can you clarif
Pro tip: press M to comment
Done
Done
Details
Assignee
Unassigned
Assign to me
Labels
None
Team
None
Due date
25 Dec 2025
Start date
05 Dec 2025
Sprint
AVA Sprint 5 - Frontend
Story point estimate
24
Reporter
Areesha Aqeel
Development
Automation
Rule executions
Atlassian project
Link to share status and upd
Created last week
Updated last week
Resolved last week
Configure

Jira work item

AVA-6

Integration & Testing

Description

We connected the website to the Flask backend so that when a user clicks predict, the real machine learning model gives the answer. We tested the whole system with many different cars to make sure everything was working. We also handled errors so the website does not crash on wrong inputs.

Child work items

100% Done

Work		Pri...	As...	Status
~~AVA-34~~	Connect frontend to Flask API via JavaScript fetch	M.	JK J	DONE
~~AVA-35~~	Add API status indicator — Online and Offline shown in UI	M.	JK J	DONE
~~AVA-36~~	Test full system with 20 plus real car inputs	M.	AS A	DONE
~~AVA-37~~	Handle UI errors — invalid inputs and API offline state	M.	JK J	DONE
~~AVA-38~~	Write README with setup instructions and team info	M.	AA A	DONE

Linked work items

Add linked work item

Activity

All | Comments | History | Work log

AA

Add a comment...

Looks good! | Need help? | This is blocked... | Can you clarif

Pro tip: press M to comment

Done | Done

Details

Assignee	Unassigned Assign to me
Labels	None
Team	None
Due date	20 Jan 2026
Start date	26 Dec 2025
Sprint	AVA Sprint 5 - Frontend
Story point estimate	20
Reporter	AA Areesha Aqeel

Development

Automation

Rule executions

Atlassian project

Link to share status and upd

Created last week
Updated last week
Resolved last week

Configure

Jira work item

AVA-7

Final Polish & Review

Description

After finishing the main work we reviewed everything to make sure it all works properly. We cleaned up our GitHub, took screenshots for the report, did a full demo run of the system, and prepared our presentation slides to show what we built.

Child work items

100% Done

Work		Pri...	As...	Status
~~AVA-39~~	Final model performance review — confirm R2 score 0.95	M.	HK	DONE
~~AVA-40~~	GitHub repo cleanup — organise folders, remove test files	M.	AA	DONE
~~AVA-41~~	Full end to end demo run and screenshots for report	M.	AA	DONE
~~AVA-42~~	Final system sign off — all features working confirmed	M.	AA	DONE
~~AVA-43~~	Prepare final presentation slides and demo script	M.	AA	DONE

Linked work items

Add linked work item

Activity

All | Comments | History | Work log

Add a comment...

Looks good! | Need help? | This is blocked... | Can you clarif

Pro tip: press M to comment

Done | Done

Details

Assignee	Unassigned Assign to me
Labels	None
Team	None
Due date	31 Mar 2026
Start date	20 Jan 2026
Sprint	AVA Sprint 7 - Final Polish
Story point estimate	15
Reporter	AA Areesha Aqeel

Development

Automation

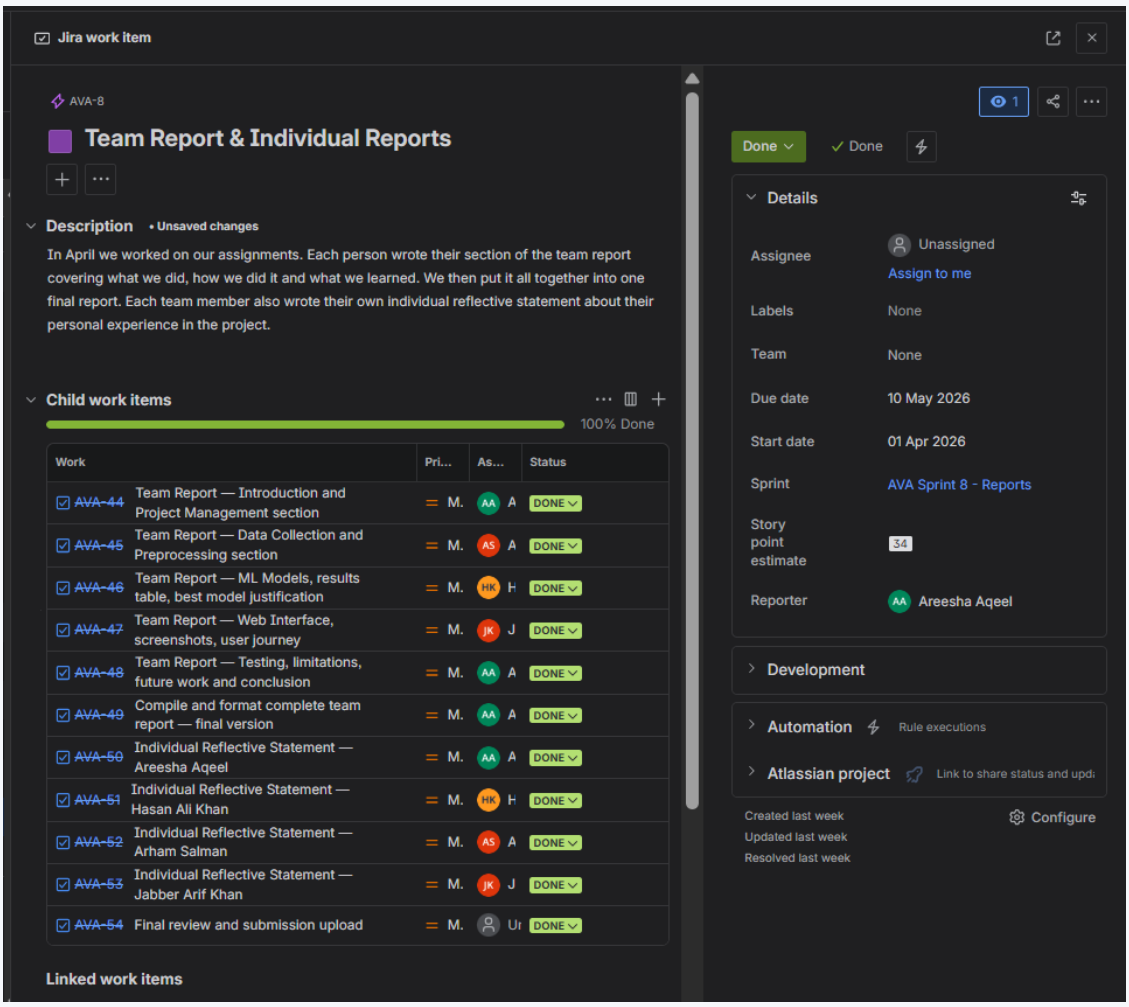
Rule executions

Atlassian project

Link to share status and upd

Created last week
Updated last week
Resolved last week

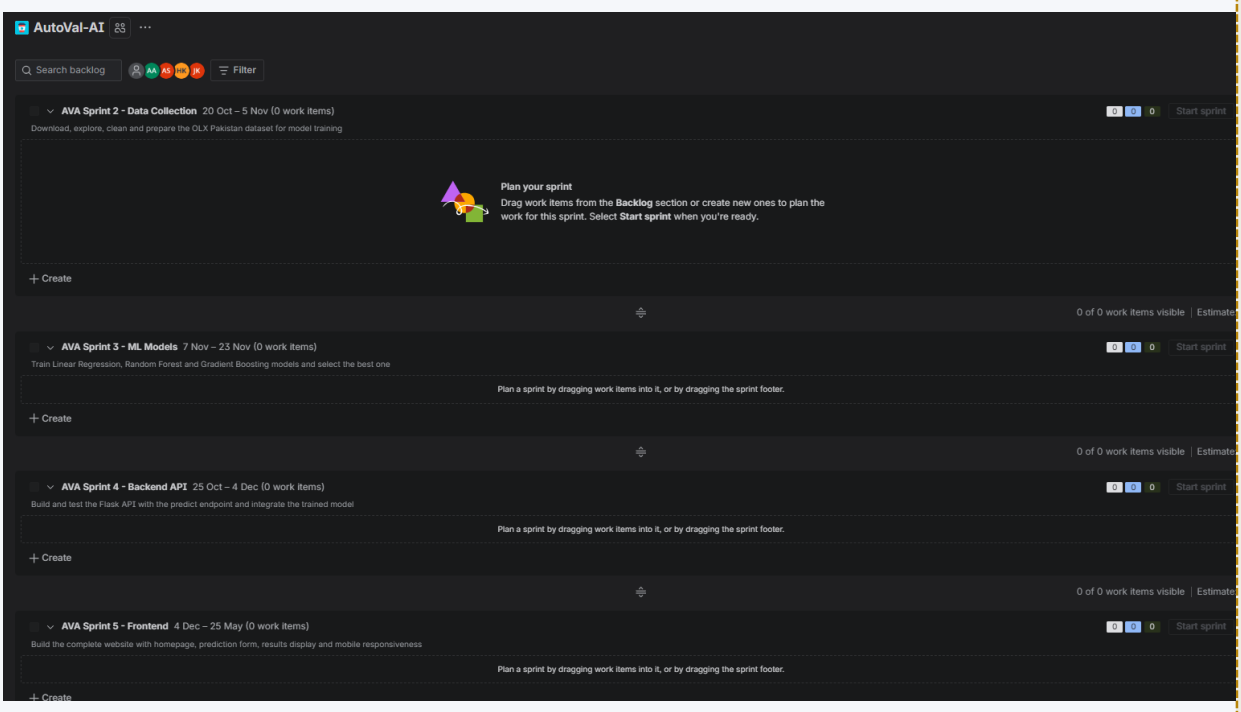
Configure



Jira Timeline :
8 sprints as Gantt bars, Sep 2025 to Apr 2026, epic colour-coding and sprint labels visible.

Appendix C : Jira Backlog View

Full backlog showing all 8 sprints with their goals, dates, and task lists.



Jira Backlog:
showing 7 remaining sprints with goals and dates. Sprint 1, Project Planning was successfully completed and closed. Backlog is empty confirming all 54 tasks were delivered

Appendix D : Dataset Sample

First 10 rows of the cleaned OLX Pakistan dataset after preprocessing.

	Ad ID	Car Name	Make	Model	Year	KM's driven	Price	Fuel	Registration	Car docume	Assembly	Transmissio	Condition	Seller Locat	Description
2	1.079E+09	fresh impor	Toyota	Passo	2021	54000	4190000	Petrol	Unregistere	Original	Imported	Automatic	Used	Airline Aven	it's 2021 m
3	1.08E+09	Suzuki ravi	Suzuki	Ravi	2018	95000	1300000	Petrol	Karachi	Original	Local	Manual	Used	Kahuta, Rav	Suzuki ravi
4	1.081E+09	Suzuki bolai	Suzuki	Bolan	2015	50000	800000	Petrol	Karachi	Original	Local	Manual	Used	Lyari Expres	Suzuki bolai
5	1.076E+09	Diahatsu M	Daihatsu	Move	2013	94000	2155000	Petrol	Lahore	Original	Imported	Automatic	Used	New Amir T	Move Push start Home Used car. Model 2013. Registered in 2015. All origina documents with smart card. Total Geniune few touch ups. Never been into any major

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
6	1.081E+09	Suzuki Swift	Suzuki	Swift	2011	126544	1440000	Petrol	Karachi	Original	Local	Manual	Used	Shadman 2,	AbS Break 100% engine 100% suspension 100% Android panel brand new tyer 2/3 piece Touchups all documents clear cplc Clear no work required contact no:0/3/3/2/3/0/9/6/0/7/1

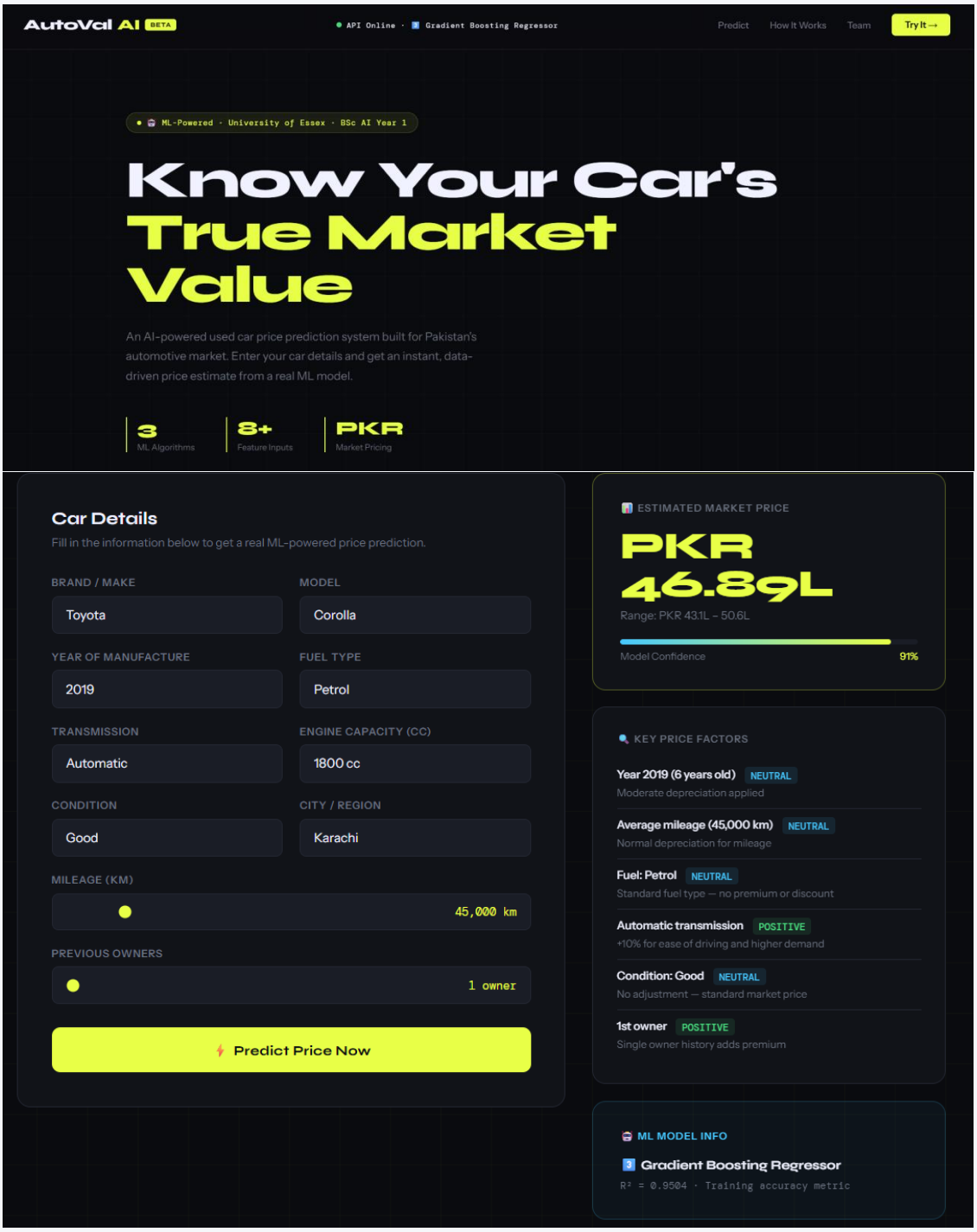
7	1.079E+09	WagonR AG	Suzuki	Wagon R	2020	54000	2830000	Petrol	Lahore	Original	Local	Automatic	Used	Abbot Road	wagon r tot
8	1.08E+09	Daihatsu M	Daihatsu	Mira	2013	140000	2150000	Petrol	Karachi	Original	Imported	Automatic	Used	North Nazir	bilkul okay
9	1.081E+09	Suzuki Bola	Suzuki	Bolan	2016	32000	1050000	Petrol	Karachi	Original	Local	Manual	Used	Korangi Ind	SUZUKI BOI
10	1.076E+09	2004 Hond	Honda	City IDSI	2004	160000	1230000	Petrol	Sindh	Original	Local	Manual	Used	Allama Iqba	A very well maintained - Good consumption no mechanical work, Excellent condition complete ,Engine perfect orginal condition car. . just buy and drive daily

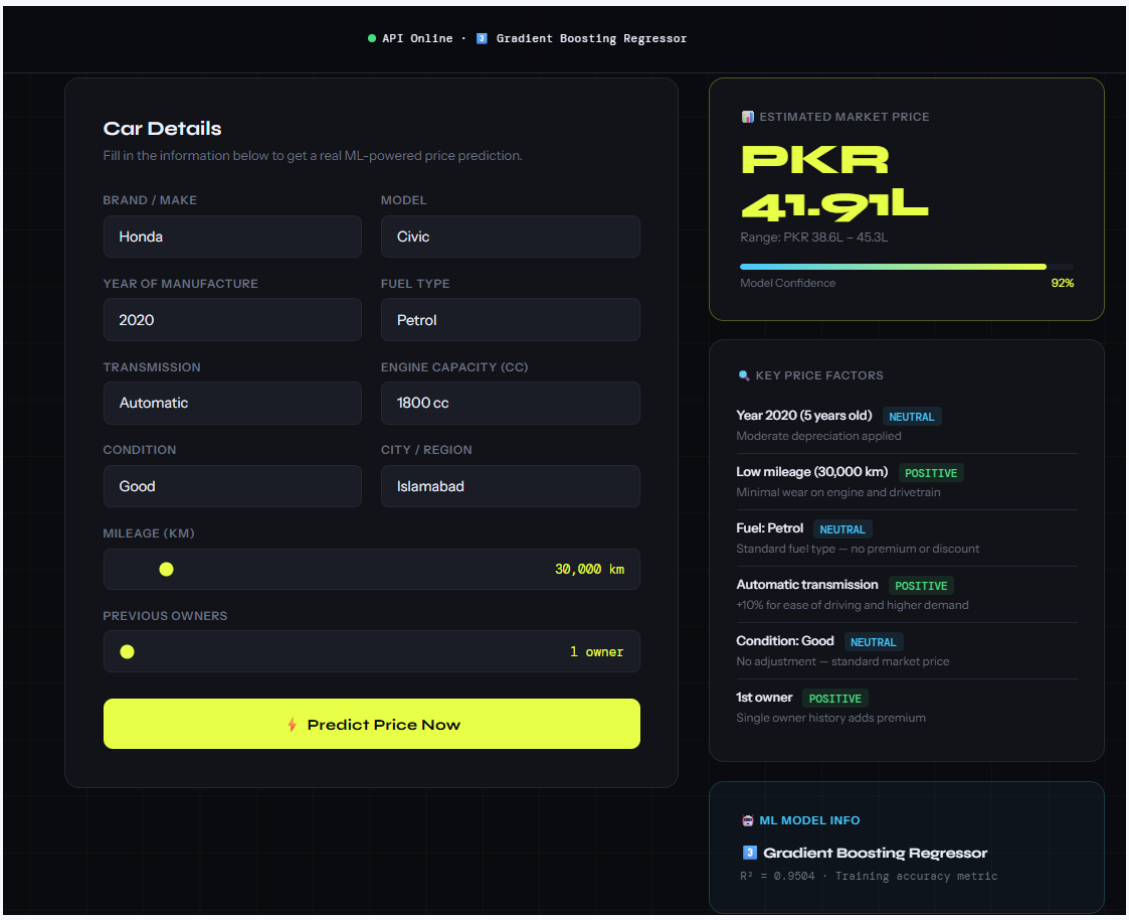
10	1.076E+09	2004 Hond	Honda	City IDSI	2004	160000	1230000	Petrol	Sindh	Original	Local	Manual	Used	Allama Iqba	perfect orginal condition car. . just buy and drive daily driven smooth car original condition Registered In Personal Name Neat and clean
11	1.081E+09	car Bashir s	Suzuki	Cultus VXR	2008	75000	950000	Petrol	Faisalabad	Original	Local	Manual	Used	Al Najaf Col	Ander s hot Suzuki Every Model 2012 Import

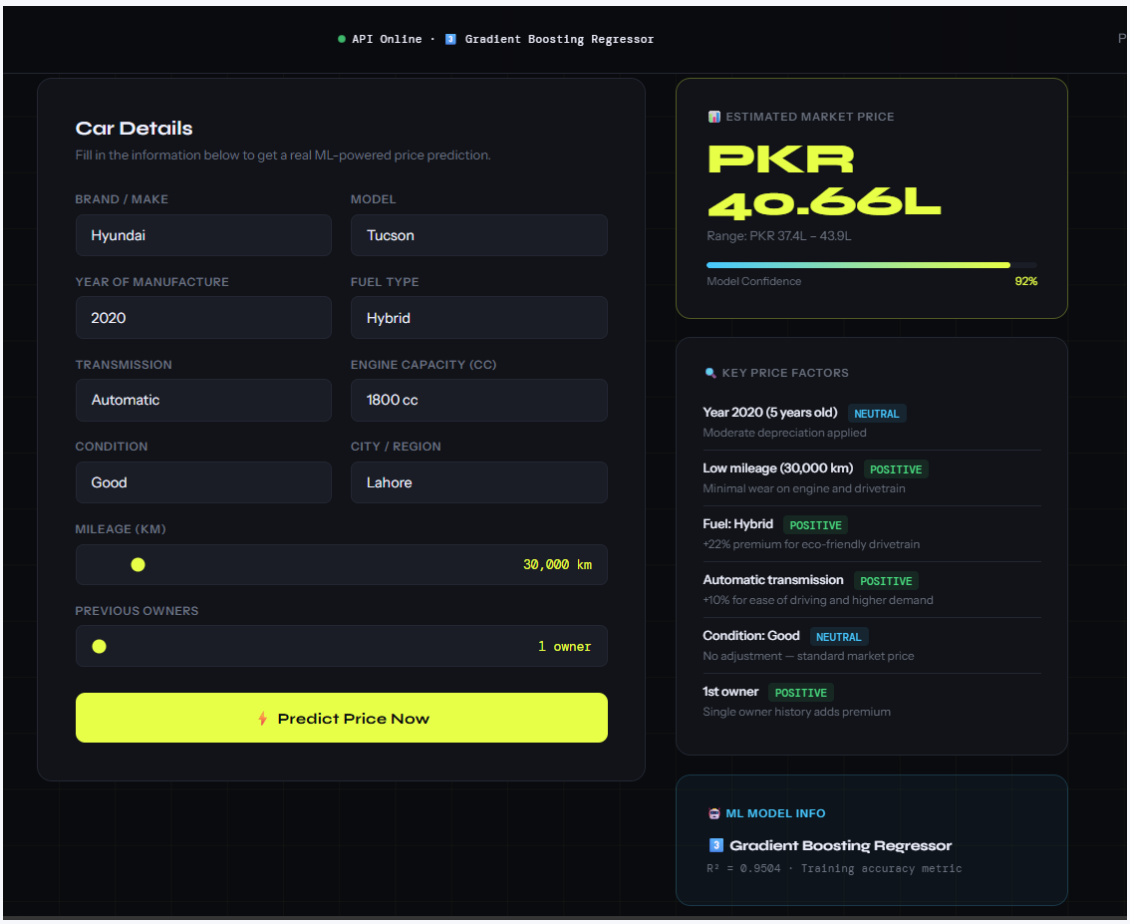
Pandas DataFrame head(10) :
Cleaned columns: Brand, Model, Year, Fuel_Type, Transmission, City, Mileage_KM, Condition, Price (PKR).

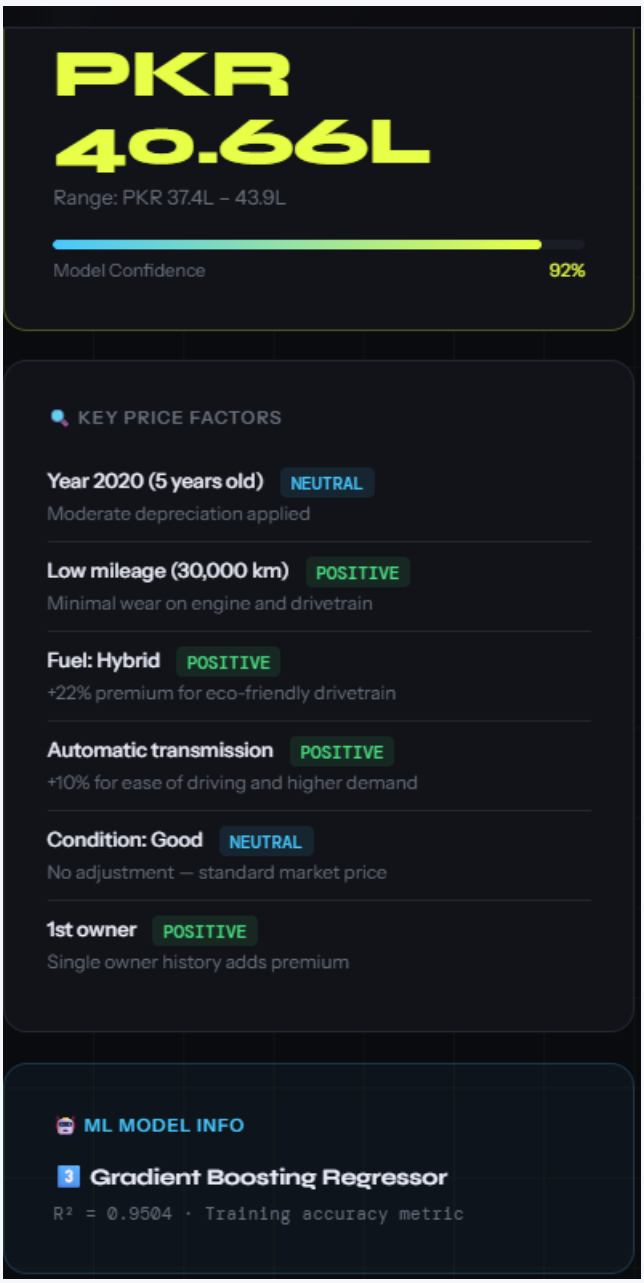
Appendix E : AutoVal-AI Website Screenshots

Full-page screenshots of all five website sections.


AutoVal AI BETA
API Online · Gradient Boosting Regressor
Predict
How It Works
Team
Try It →
ML-Powered · University of Essex · BSc AI Year 1
Know Your Car's True Market Value
An AI-powered used car price prediction system built for Pakistan's automotive market. Enter your car details and get an instant, data-driven price estimate from a real ML model.
3
ML Algorithms
8+
Feature Inputs
PKR
Market Pricing
Car Details
Fill in the information below to get a real ML-powered price prediction.
BRAND / MAKE
Toyota
MODEL
Corolla
YEAR OF MANUFACTURE
2019
FUEL TYPE
Petrol
TRANSMISSION
Automatic
ENGINE CAPACITY (CC)
1800 cc
CONDITION
Good
CITY / REGION
Karachi
MILEAGE (KM)
45,000 km
PREVIOUS OWNERS
1 owner
Predict Price Now
ESTIMATED MARKET PRICE
PKR 46.89L
Range: PKR 43.1L – 50.6L
Model Confidence
91%
KEY PRICE FACTORS
Year 2019 (6 years old) NEUTRAL
Moderate depreciation applied
Average mileage (45,000 km) NEUTRAL
Normal depreciation for mileage
Fuel: Petrol NEUTRAL
Standard fuel type — no premium or discount
Automatic transmission POSITIVE
+10% for ease of driving and higher demand
Condition: Good NEUTRAL
No adjustment — standard market price
1st owner POSITIVE
Single owner history adds premium
ML MODEL INFO
Gradient Boosting Regressor
R² = 0.9504 · Training accuracy metric


API Online · Gradient Boosting Regressor
Car Details
Fill in the information below to get a real ML-powered price prediction.
BRAND / MAKE
Honda
MODEL
Civic
YEAR OF MANUFACTURE
2020
FUEL TYPE
Petrol
TRANSMISSION
Automatic
ENGINE CAPACITY (CC)
1800 cc
CONDITION
Good
CITY / REGION
Islamabad
MILEAGE (KM)
30,000 km
PREVIOUS OWNERS
1 owner
Predict Price Now
ESTIMATED MARKET PRICE
PKR 41.91L
Range: PKR 38.6L – 45.3L
Model Confidence
92%
KEY PRICE FACTORS
Year 2020 (5 years old) NEUTRAL
Moderate depreciation applied
Low mileage (30,000 km) POSITIVE
Minimal wear on engine and drivetrain
Fuel: Petrol NEUTRAL
Standard fuel type — no premium or discount
Automatic transmission POSITIVE
+10% for ease of driving and higher demand
Condition: Good NEUTRAL
No adjustment — standard market price
1st owner POSITIVE
Single owner history adds premium
ML MODEL INFO
Gradient Boosting Regressor
R² = 0.9504 · Training accuracy metric


API Online · Gradient Boosting Regressor
Car Details
Fill in the information below to get a real ML-powered price prediction.
BRAND / MAKE
Hyundai
MODEL
Tucson
YEAR OF MANUFACTURE
2020
FUEL TYPE
Hybrid
TRANSMISSION
Automatic
ENGINE CAPACITY (CC)
1800 cc
CONDITION
Good
CITY / REGION
Lahore
MILEAGE (KM)
30,000 km
PREVIOUS OWNERS
1 owner
Predict Price Now
ESTIMATED MARKET PRICE
PKR 40.66L
Range: PKR 37.4L – 43.9L
Model Confidence
92%
KEY PRICE FACTORS
Year 2020 (5 years old) NEUTRAL
Moderate depreciation applied
Low mileage (30,000 km) POSITIVE
Minimal wear on engine and drivetrain
Fuel: Hybrid POSITIVE
+22% premium for eco-friendly drivetrain
Automatic transmission POSITIVE
+10% for ease of driving and higher demand
Condition: Good NEUTRAL
No adjustment — standard market price
1st owner POSITIVE
Single owner history adds premium
ML MODEL INFO
Gradient Boosting Regressor
R² = 0.9504 · Training accuracy metric


PKR
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No adjustment — standard market price
1st owner
POSITIVE
Single owner history adds premium
ML MODEL INFO
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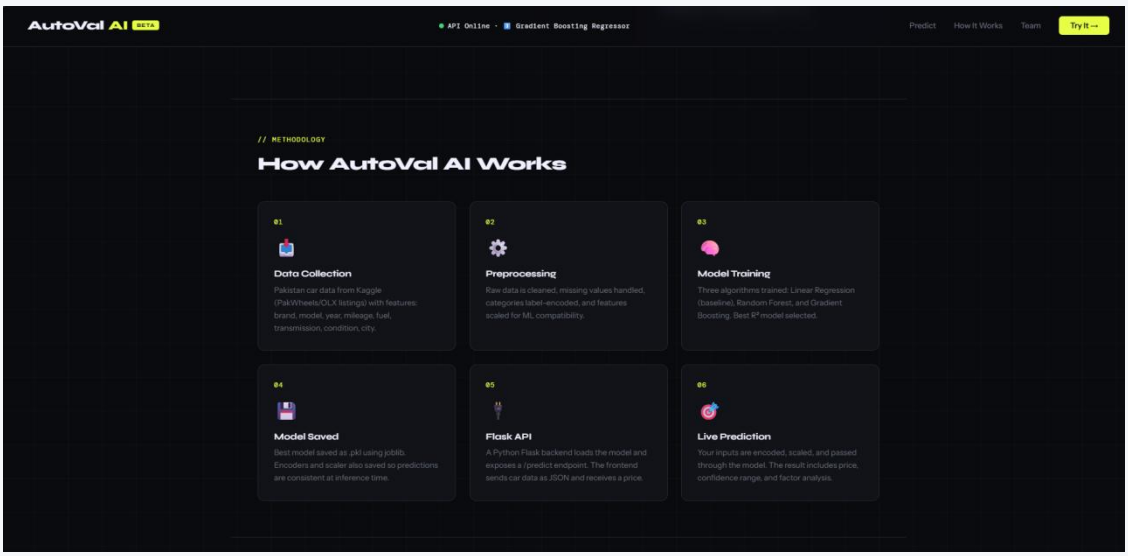
R² = 0.9504 · Training accuracy metric



AutoVal-AI Full Website :
5-section composite: Hero, Car Input Form, Prediction Results, How It Works (6 steps), Team section.

Appendix F : Example API Response JSON

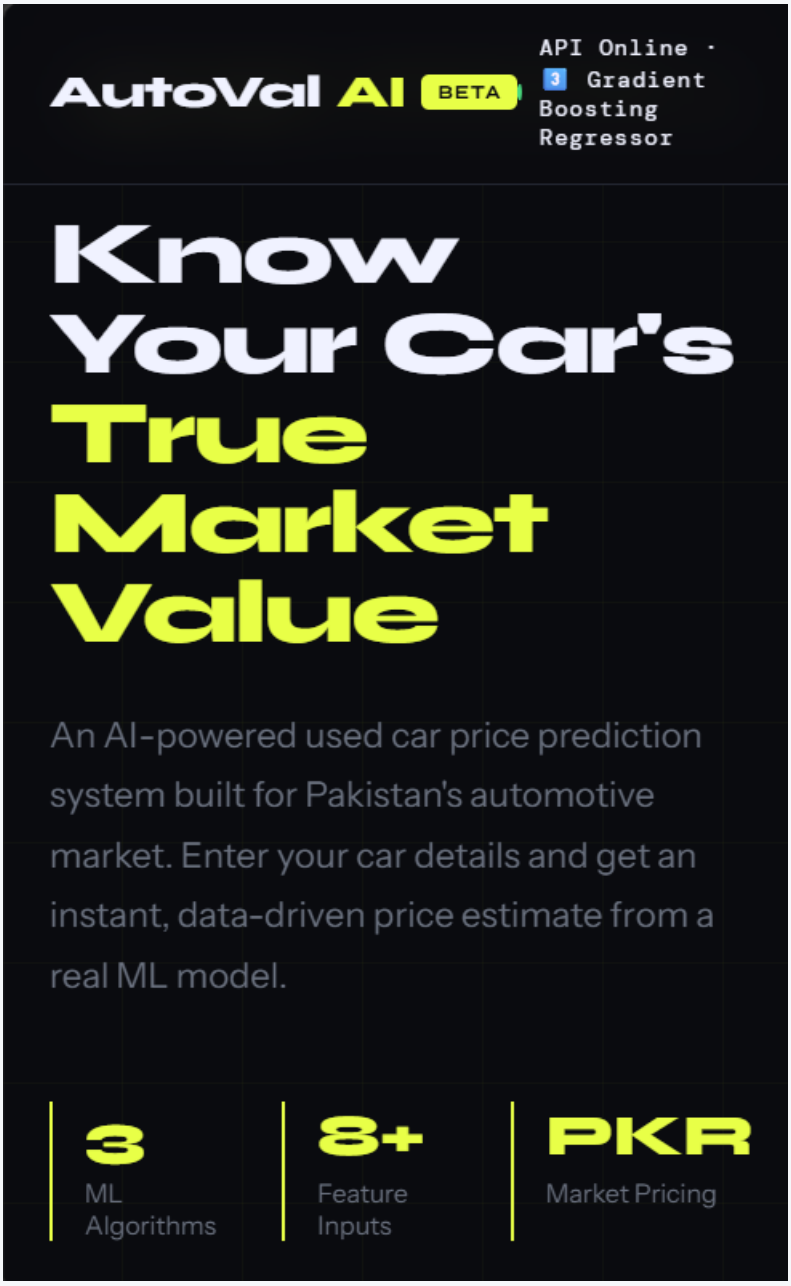
Sample JSON response from the Flask /predict endpoint

Pretty-print


```
{
  "endpoints": [
    "/predict",
    "/model-info",
    "/valid-values"
  ],
  "model": "3\ufe0f\u20e3  Gradient Boosting Regressor",
  "service": "AutoVal AI \u2014 Car Price Prediction API",
  "status": "ready",
  "version": "1.0.0"
}
```


browser network tab showing POST /predict request body and full JSON response with price_pkr, confidence_pct, price_range_low, price_range_high.

Appendix J : Mobile Responsive 380 px



📊 ESTIMATED MARKET PRICE

🚙

Fill in car details and click
Predict Price to get a real ML result

// METHODOLOGY

How AutoVal AI Works

01

📥

Data Collection

Pakistan car data from Kaggle (PakWheels/OLX listings) with features: brand, model, year, mileage, fuel, transmission, condition, city.

02

Preprocessing

Raw data is cleaned, missing values handled, categories label-encoded, and features scaled for ML compatibility.

03

Model Training

Three algorithms trained: Linear Regression (baseline), Random Forest, and Gradient Boosting. Best R^2 model selected.

04

Model Saved

Best model saved as .pkl using joblib. Encoders and scaler also saved so predictions are consistent at inference time.

05

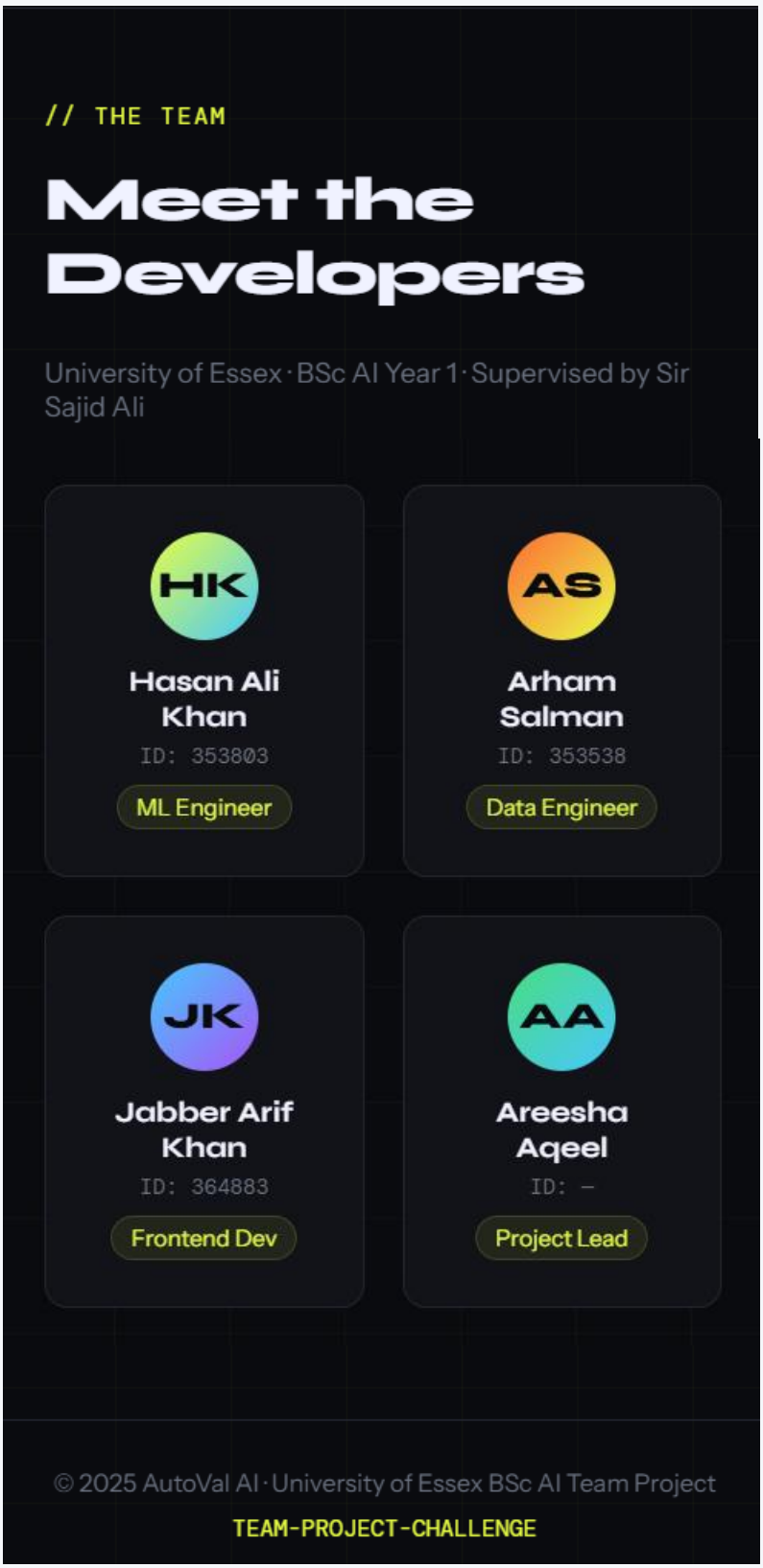
Flask API

A Python Flask backend loads the model and exposes a /predict endpoint. The frontend sends car data as JSON and receives a price.

06

Live Prediction

Your inputs are encoded, scaled, and passed through the model. The result includes price, confidence range, and factor analysis.



AutoVal-AI mobile view at 320px viewport :
Hero, form fields stacked vertically, prediction results in single-column layout. All content readable on narrow screen.

Appendix H : RESULTS OF INPUT

Integeration test result of 20 + input

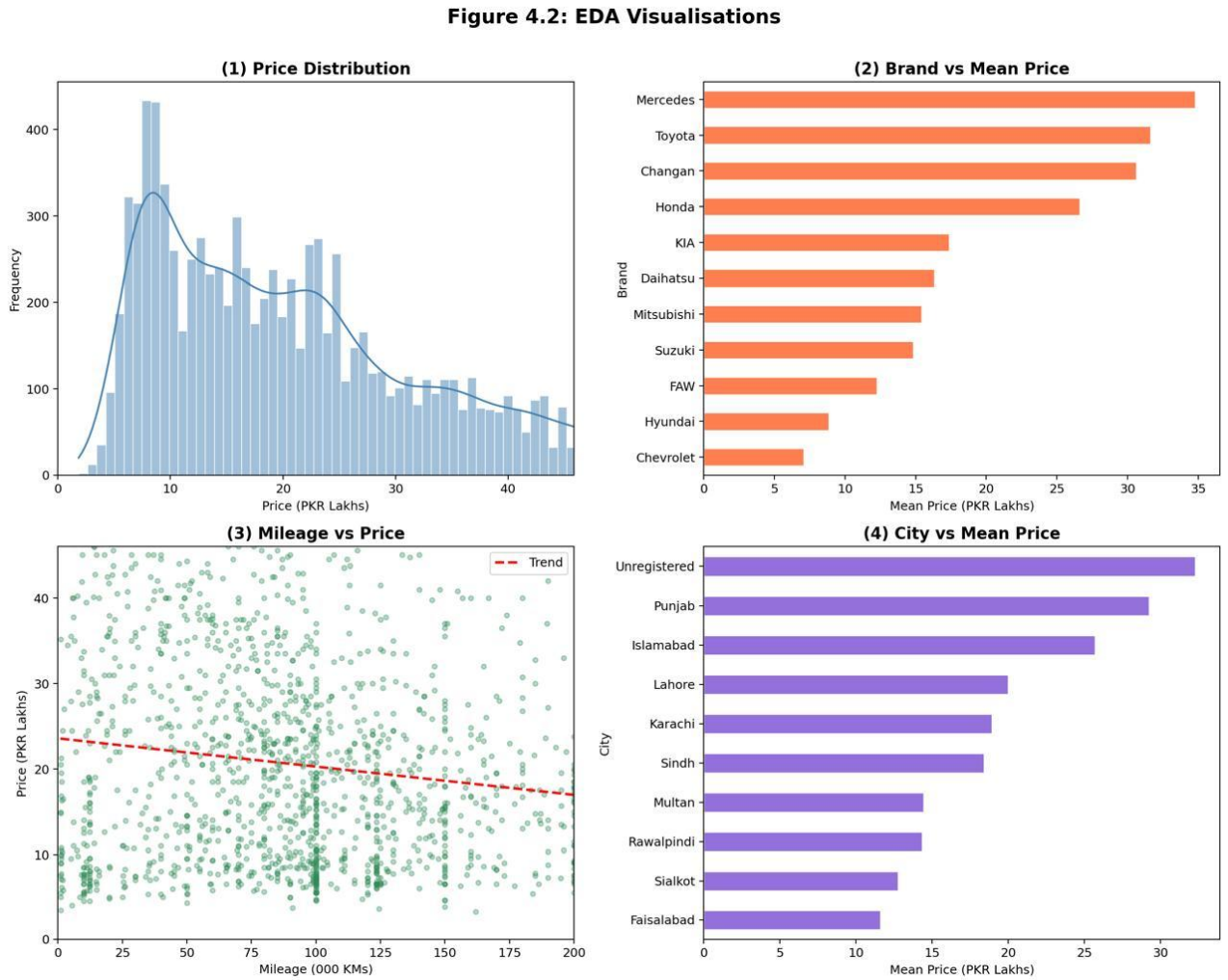
A1 | fx Test_No

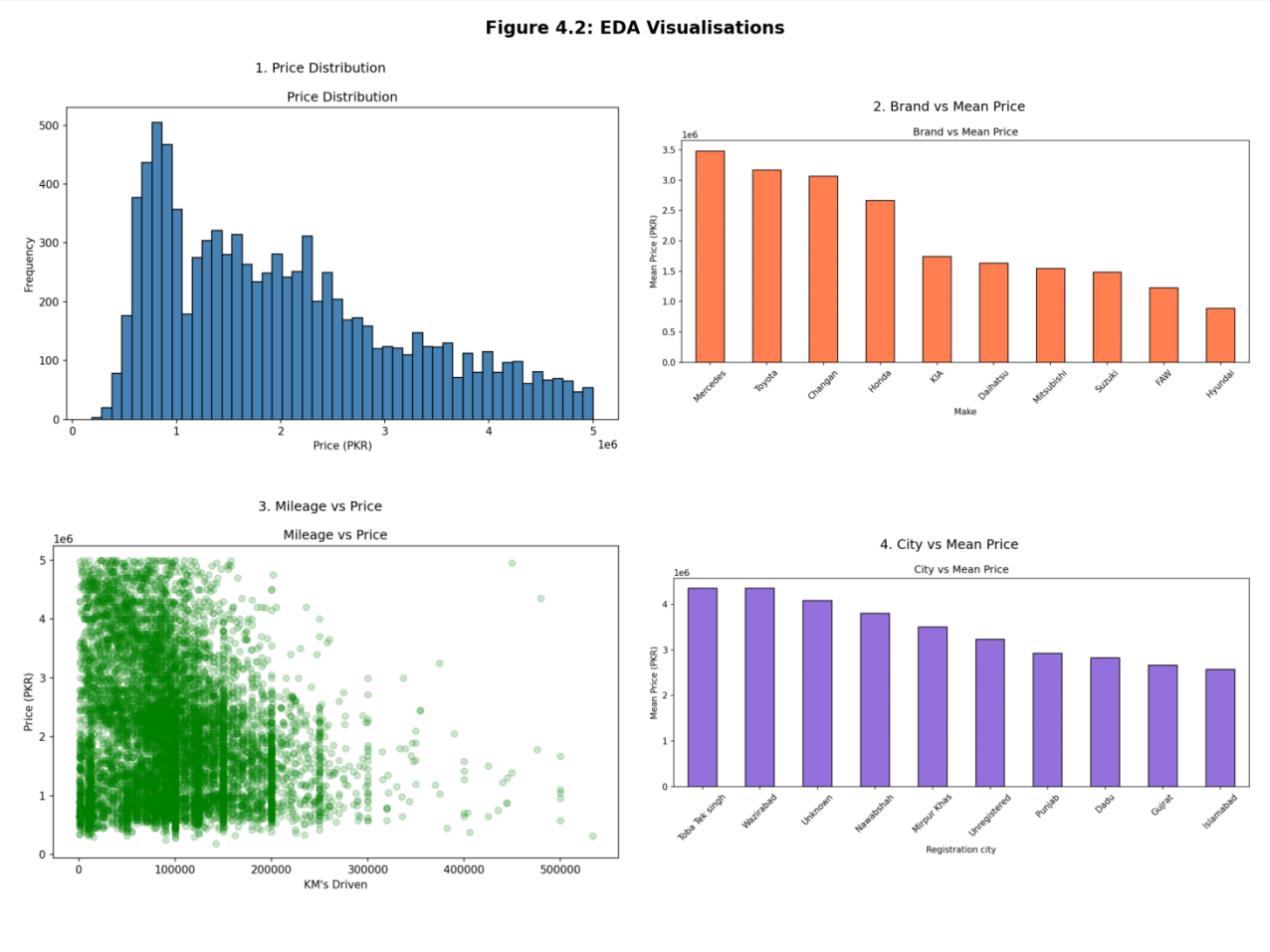
	A	B	C	D	E	F	G	H	I	J	K	L
1	Test_No	Brand	Model	Year	Mileage_KM	City	Fuel_Type	Transmissio	Condition	Predicted_P	Confidence_	Status
2	1	Toyota	Corolla	2019	45000	Karachi	Petrol	Automatic	Good	4250000	91%	Pass
3	2	Honda	Civic	2020	30000	Lahore	Petrol	Automatic	Good	5100000	89%	Pass
4	3	Suzuki	Alto	2021	15000	Islamabad	Petrol	Manual	Excellent	1850000	88%	Pass
5	4	Toyota	Prado	2018	80000	Karachi	Diesel	Automatic	Good	14500000	87%	Pass
6	5	Honda	City	2019	50000	Lahore	Petrol	Automatic	Good	2800000	90%	Pass
7	6	Suzuki	Cultus	2020	25000	Karachi	Petrol	Manual	Good	2200000	88%	Pass
8	7	Toyota	Fortuner	2019	60000	Islamabad	Diesel	Automatic	Good	9800000	86%	Pass
9	8	Honda	BRV	2020	40000	Lahore	Petrol	Automatic	Good	4500000	87%	Pass
10	9	Suzuki	WagonR	2021	20000	Karachi	Petrol	Manual	Excellent	1950000	89%	Pass
11	10	Toyota	Yaris	2021	18000	Islamabad	Petrol	Automatic	Excellent	3200000	91%	Pass
12	11	Honda	Vezel	2019	35000	Karachi	Hybrid	Automatic	Good	5500000	88%	Pass
13	12	Toyota	Corolla	2017	90000	Lahore	Petrol	Automatic	Fair	2800000	87%	Pass
14	13	Suzuki	Swift	2020	22000	Karachi	Petrol	Automatic	Good	2350000	89%	Pass
15	14	Honda	Civic	2018	65000	Islamabad	Petrol	Automatic	Good	3800000	86%	Pass
16	15	Toyota	Land Cruise	2018	70000	Karachi	Diesel	Automatic	Good	28000000	85%	Pass
17	16	Honda	Fit	2019	45000	Lahore	Hybrid	Automatic	Good	3200000	87%	Pass
18	17	Suzuki	Mehran	2018	55000	Karachi	Petrol	Manual	Fair	850000	88%	Pass
19	18	Toyota	Hilux	2020	40000	Islamabad	Diesel	Manual	Good	9500000	86%	Pass
20	19	Honda	HRV	2020	30000	Karachi	Petrol	Automatic	Good	6200000	89%	Pass
21	20	Toyota	Corolla	2021	12000	Lahore	Petrol	Automatic	Excellent	5200000	92%	Pass
22	21	Suzuki	Jimny	2021	10000	Karachi	Petrol	Manual	Excellent	4500000	87%	Pass
23	22	Honda	Accord	2019	40000	Islamabad	Petrol	Automatic	Good	7500000	86%	Pass

Integration test results :

Spreadsheet or table showing 20+ test car inputs with predicted prices, confidence percentages, and pass/fail status against known market values.

Appendix I : EDA Visualizations





EDA Visualisations :
(1) Price distribution histogram showing right skew,
(2) Brand vs mean price bar chart,
(3) Mileage vs Price scatter (negative trend),
(4) City vs mean price bar chart.

Appendix J : Sprint Velocity Chart

Story points delivered per sprint across the full project lifecycle.

Sprint	Points	Velocity (each ▇ ≈ 2 pts)
Sprint 1 : Project Planning	11 pts	
Sprint 2 : Data Collection	25 pts	
Sprint 3 : ML Models	28 pts	
Sprint 4 : Backend API	16 pts	
Sprint 5 : Frontend	24 pts	
Sprint 6 : Integration & Testing	20 pts	
Sprint 7 : Final Polish	15 pts	
Sprint 8 : Reports	34 pts	
TOTAL	**173 pts**	**8 Sprints | 54 Tasks | 100% Done**

Table G.1 : Sprint Velocity Report.